Energy from the Sun

Exam code: 0625 & 0972

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The Sun

Extended tier only

- The Sun transfers energy to Earth by **electromagnetic radiation**
- Most of our energy resources on Earth come indirectly from the **Sun**:
  - The Sun heats up the atmosphere, creating **wind** and producing **waves**
  - Water evaporated by the Sun falls as rain, filling up **reservoirs**
  - Plants grown using sunlight form the basis for fuels – both biofuels and fossil fuels
- Some energy resources, however, do not come from the Sun
- These include:
  - **Geothermal** – this comes from heat produced in the Earth's core
  - **Nuclear** – this comes from elements which make up a small proportion of the Earth's crust
  - **Tidal** – this comes (mainly) from the gravitational attraction of the Moon

The Sun

Energy from the Sun is the source of the energy stored in many of Earth's energy resources

Stuck on something?

Ask questions and get help with exactly what you're revising.

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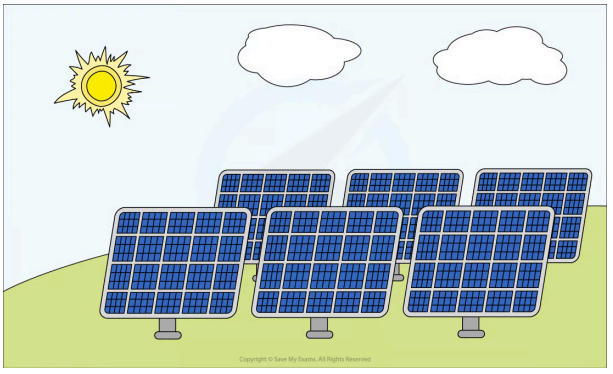
Solar cells

- The energy from the Sun that falls on the Earth is transferred by radiation
  - Mostly **visible light** and **infrared radiation**
- The amount of energy transferred from the Sun to the Earth **each hour** is equal to the energy use of the world for **one year!**
- Therefore, scientists are working hard to find methods of harnessing this energy
  - Solar energy has a **low energy density**, which means large collecting devices are required
  - Collecting solar energy has been **expensive** (due to the equipment required) and **inefficient**
  - Although **improved technology** has improved efficiency, **greater demand** has lowered costs
- **Solar cells** transfer energy from sunlight and produce a current, therefore generating **electricity**
  - Solar cells, sometimes called photovoltaic cells, are made of semiconducting materials
  - A number of cells connected together can supply electricity to homes, small-scale businesses, communication devices and satellites
  - Energy generated from solar cells can be stored in batteries for later use

Solar cells



Solar cells use energy from sunlight to produce electricity

    - A **renewable** resource is one which can be **replenished at the same rate as it is used**

- In many places on Earth sunlight is a **reliable** energy resource (this means that the sun shines most of the time)
- Solar farms produce no **greenhouse gases** or **pollution**
- Solar energy can be generated in **remote places** where they don't have electricity
    - For example to power solar street signs in rural areas

- Solar cells are readily available
- Solar farms are relatively quick and cheap to build, and inexpensive to run compared to power stations
- Solar cells can be placed on top of buildings

Disadvantages of solar cells

- Solar farms need to be **large scale** to produce large amounts of electricity
- In many places on Earth, sunlight is **not** a **reliable** energy resource (there are not enough sunshine-hours to justify the set-up costs)
- Solar cells remain fairly expensive for households to install and maintain (they must be kept clean to work at full capacity)
- Each solar cell only produces a small amount of electricity so solar farms are needed to produce large amounts of electricity
- Solar cells are relatively inefficient (around 20% - 40%)

Solar panels

- Useful energy may be obtained from infrared and other electromagnetic waves from the Sun using **solar panels**

- **Solar panels** transfer energy from sunlight to the thermal store of the solar panels which is used to **heat water** in the pipes
    - Solar panels can be used to warm domestic water supplies
        - This can reduce the cost of producing hot water since it is heated partially by the solar panels

    - **Solar furnaces** consist of large curved mirrors that focus the sun's rays on to a small area
        - These can be used to boil water, generating enough steam to turn turbines and generate electricity in a power station

Solar panel

Solar panels use energy from sunlight to heat water

- Water is **stored** in a water **tank** and pumped through small **pipes** running through the solar panel
- I**nfrared radiation** from the Sun heats the water in the pipes as it flows through them
- The water returns to the tank at a **higher temperature**
- Water may still need to be heated further to reach the desired temperature (eg. for a shower)

Advantages of solar panels

- Solar energy is a **renewable** resource
- In many places on Earth, sunlight is a **reliable** energy resource (this means that the sun shines most of the time)
- Solar panels produce no **greenhouse gases** or **pollution** once they are operating
- Solar panels can cut the cost of energy bills for households
- Solar panels can be placed on the roofs of buildings

Disadvantages of solar panels

- Solar furnaces need to be large scale to produce high temperatures
- Energy is still needed to heat water to a higher temperature in domestic households
- In many places on Earth, sunlight is **not** a **reliable** energy resource (the sun doesn't shine regularly enough to justify the set-up costs)

Wind energy

- Energy from the **wind** can be used to generate **electricity**
- The wind turns a **turbine**, which turns a **generator**, which generates electricity
- Energy from the kinetic store of the wind is transferred
  - mechanically to the kinetic store of the turbine
  - which is transferred mechanically to the kinetic store of the generator

Wind turbines use energy from the wind to generate electricity

Advantages of wind power

- Wind is a **renewable** resource
- Wind turbines produce **no** greenhouse gases or pollution
- Wind farms are relatively **inexpensive** to build and cheap to run
- Wind turbines have a relatively high **efficiency** of 50%
- The land used for wind farms can still be used for **farming**
- **Off-shore** wind farms can be built in coastal areas
- Wind turbines do not use fuel, so there are no fuel costs

Disadvantages of wind power

- Wind power is **not reliable** because the wind does not blow constantly
- Turbines are built facing a specific direction so they only work when the wind is blowing in that specific direction
- Some people consider wind farms to be **visual pollution** , so there can be local opposition to the construction of wind farms in some towns and cities
- Each turbine only produces a small amount of electricity, so wind farms are needed for large scale electricity generation
- High-voltage power lines are required to transmit the electricity to the National Grid

Examiner Tips and Tricks

In your IGCSE exam, you may be required to discuss the advantages and disadvantages of energy resources. You may also need to apply your understanding of energy resources to unfamiliar scenarios, for example, by suggesting which type

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